File permissions in Linux

Project description

In this project, I used Linux commands to inspect and manage file and directory permissions in a research team's workspace. I identified permissions that did not comply with the organization's security policies and used chmod to restrict unauthorized access. This activity demonstrates the importance of access control and the principle of least privilege in cybersecurity.

Check file and directory details

I used the following commands to inspect the files and directories, including hidden files:

```
cd /home/researcher2/projects
ls -la
```

→ The `ls -la` command displays all files and directories, including hidden files, along with their permissions, owner, group, size, and modification date.

The current permissions are:

File or directory	Permission string
project_k.txt	-rw-rw-rw-
project_m.txt	-rw-r-----
project_r.txt	-rw-rw-r--
project_t.txt	-rw-rw-r--
.project_x.txt	-rw--w----
drafts	drwx--x---

Describe the permissions string

A Linux permission string contains 10 characters. The first character identifies the file type: - indicates a regular file and d indicates a directory. The next three characters represent the owner's permissions, the following three represent the group's permissions, and the last three represent permissions for others.

The characters r, w, and x represent read, write, and execute permissions, respectively. A hyphen (-) means that the permission is not granted.

For example, -rw-rw-rw- represents a regular file for which the owner, group, and others all have read and write permissions, but nobody has execute permission. This violates the organization's policy because others have write access.

Change file permissions

The file project_k.txt grants write permission to others, which violates the organization's security policy. I removed write permission for others using: `chmod o-w project_k.txt`

The chmod command changes file permissions. The option o-w removes write permission from others without changing the owner's or group's permissions.

I verified the change using: `ls -l project_k.txt`

The expected permission string after the change is -rw-rw-r--. Others can still read the file, but they can no longer modify it

Change file permissions on a hidden file

The hidden file .project_x.txt initially grants write permission to both the owner and the group. I changed the permissions to allow read-only access for the owner and group and no access for others.

I used the following command: `chmod ug=r,o= .project_x.txt`

The ug=r option gives the owner and group read-only permission, while o= removes all permissions from others.

I verified the change using: `ls -l .project_x.txt`

The expected permission string is -r--r-----. This configuration prevents writing by all users while allowing the owner and group to read the file.

Change directory permissions

The drafts directory initially grants execute permission to the group, which means group members may be able to access the directory even though they cannot list its contents. The organization's policy requires that only researcher2, the owner, can access the directory.

I used the following command: `chmod 700 /home/researcher2/projects/drafts`

I verified the change using: `ls -ld /home/researcher2/projects/drafts`

The expected permission string is drwx------. This restricts access to the directory to its owner.

Summary

I used ls -la to inspect file and directory permissions, including hidden files, and learned to interpret Linux permission strings. I used chmod to remove unauthorized write access from project_k.txt, configure read-only access for .project_x.txt, and restrict the drafts directory to its owner. These actions help protect files against unauthorized modification and ensure that access is granted according to the organization's security policies.